

Measuring Five-Nines Reliability: Sample-Efficient LLM Evaluation in Saturated Benchmarks

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Abstract

While existing benchmarks demonstrate the near-perfect performance of large language models (LLMs) on various tasks, this apparent saturation often obscures the need for rigorous evaluation of their reliability. In real-world deployment, however, achieving extremely high reliability (e.g., “five-nines” (99.999%) vs. “three-nines” (99.9%)) is fundamentally critical, as this gap results in an order-of-magnitude increase in failures, which is catastrophic in reliability-critical applications. Still, estimating such a rare failure probability with tight confidence bounds requires prohibitively large LLM inference sizes, making standard Monte Carlo evaluation infeasible under limited compute budgets. In this paper, we observe that LLM failures exhibit strong systematic patterns: across broad parameterized input spaces, a small subset of inputs disproportionately accounts for the majority of failures. Leveraging this observation, we propose to learn a sampling distribution concentrated on failure-prone inputs via the cross-entropy method (CEM). We evaluate our framework on three LLMs, Qwen2.5-Math-7B-Instruct, gpt-oss-20b-low, and Gemini 2.5 Flash Lite, across parameterized GSM8K templates and achieve up to $156.22\times$ reduction in required inferences compared to naive uniform sampling. Our estimates reveal that models with indistinguishable accuracy on standard benchmarks can differ substantially in estimated failure rates, underscoring that reliability is a distinct and measurable axis of model quality. Our simple yet practical framework enables the evaluation of extreme reliability in LLMs, a distinct and underexplored dimension of evaluation beyond existing benchmarks, for their growing use in reliability-sensitive applications.

1 Introduction

Large language models (LLMs) show rapid progress not just in completing tasks with shorter duration (METR, 2026), but in giving saturating performance in numerous benchmarks (Maslej et al., 2025). For example, in AIME 2025 (American Invitational Mathematics Examination), GPT-5.2 Thinking (OpenAI, 2025) and Gemini-3-Pro (Google DeepMind, 2025) achieve a perfect score, while Claude Opus 4.6 (Anthropic, 2026) reaches 99.8%. While these near-perfect performances give the impression that models seem reliable at first glance, they may instead obscure how genuinely reliable these models are in real-world deployments where reliability becomes extremely important. As an example, in safety-critical settings such as refusing harmful requests (OpenAI, 2026; Anthropic, 2026), even a vanishingly small gap in accuracy, e.g., 99.999% (“five-nines”) and 99.9% (“three-nines”) in millions of queries, translates into an order-of-magnitude difference in failures, e.g., 1000 vs. 10 failures. Thus, this dimension of reliability evaluation is increasingly important but remains under-explored in current evaluation pipelines of existing benchmarks (Vendrow et al., 2024).

One major practical bottleneck is that, for such rare failures, achieving the tight confidence bounds requires a prohibitively large sample size via Monte Carlo sampling under a uniform

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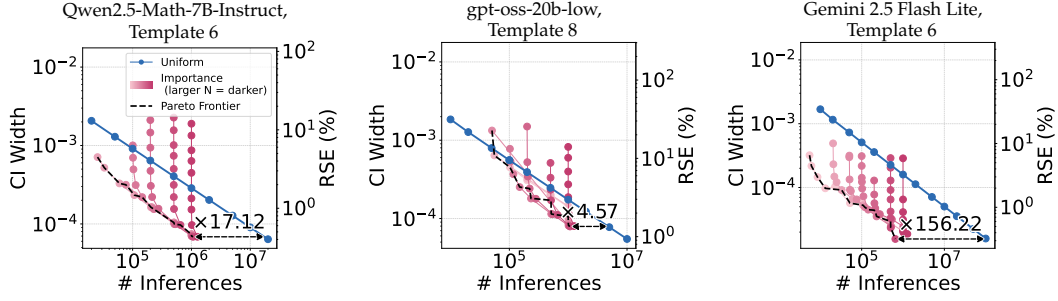


Figure 1: Each solid pink lines connects estimations with varying number of inferences (pink dots) under importance sampling with same number of overhead inferences for CEM (N), and darker pink means larger N . Similarly, solid blue line for uniform sampling. Only estimates with coverage within 1% tolerance of 99% are shown. The Pareto-frontier (dotted line) is the lower-left envelope across all $(N, \# \text{ evaluation})$ combinations, which achieves a tight CI width, i.e., relative standard error (RSE) of around 1%, at a much smaller inference budget compared to uniform sampling (e.g., $156.22\times$ gain for Gemini 2.5 Flash Lite on Template 6).

distribution P . For instance, to estimate a failure rate of $p = 10^{-5}$ (equally 99.999% accuracy) with small standard error ε , the binomial estimator $\sqrt{p(1-p)/n}$ requires $n \approx p/\varepsilon^2 = 10^{-5}/(10^{-6})^2 = 10^7$ when $\varepsilon = 10^{-6}$. Combined with the expense of LLM inference (which grows further with long reasoning traces in recent models), exhaustive uniform sampling for evaluation becomes computationally infeasible.

In this paper, we observe that *LLM failures are not uniformly distributed but exhibit systematic patterns*. Specifically, in parameterized GSM8K (Mirzadeh et al., 2025), across a combinatorially large input space, only a small set of inputs consistently induces failures at disproportionately high rates. For example, gpt-oss-20b-low (Agarwal et al., 2025) shows that approximately 82% of its failures strongly co-occur with the input prompts with a specific parameter value (e.g., fraction of blue ball = $2/7$ for GSM template 7), while other values rarely cause failures (Figure 2 middle row, Figure 3 right panel). Such concentration appears across different models, such as Qwen2.5-Math-7B-Instruct (Yang et al., 2024) and Gemini 2.5 Flash Lite (Comanici et al., 2025), and other templates in GSM. This phenomenon sharpens under self-consistency decoding with majority voting (Wang et al., 2023), which filters out random errors and reveals inputs that consistently trigger failures across multiple generations.

This systematic failure structure motivates our approach—rather than sampling uniformly, we concentrate *sampling on failure-prone regions*. We cast this as a rare-event estimation problem and adopt the CEM (Rubinstein, 1999; De Boer et al., 2005) to learn a new sampling distribution Q concentrated on failure-prone regions. From a small number of inferences, Q iteratively converges to a distribution that assigns higher probability to failure-prone input regions, which enables importance sampling for inference-efficient error estimation.

Our evaluation testbed includes multiple parameterized GSM8K templates (Cobbe et al., 2021; Mirzadeh et al., 2025), and three small but capable LLMs: Qwen2.5-Math-7B-Instruct, gpt-oss-20b-low, and Gemini 2.5 Flash Lite. Across models and templates, our method gains substantial inference efficiency, e.g., $156.22\times$ reduction in the required number of inferences for Gemini 2.5 Flash Lite on Template 6, for equivalent confidence bounds compared to uniform sampling (Figure 1 third plot). Notably, these gains grow as LLM failures become more concentrated, i.e., via increasing majority-vote ensemble size K . Specifically, efficiency improves from $3.68\times$ at $K = 4$ to $16.41\times$ at $K = 16$ for Qwen2.5-Math-7B-Instruct on Template 0 (Table 2). Our estimation results give $\sim 1\%$ relative standard errors for models with nearly 0.1% – 0.01% error rates, indicating that our method provides a practical evaluation pipeline for models’ reliability on extremely saturated tasks. Applying this pipeline across models, we find that models all exceeding 99.9% accuracy still differ by up to $2.4\times$ in estimated failure rates (e.g., gpt-oss-20b-low at 0.0254% vs. Qwen2.5-Math-7B-

Instruct at 0.0614% on Template 0, $K=16$), confirming that saturated benchmarks retain discriminative power at the reliability level (Figure 5).

2 Related Work

Evaluating reliability in saturated benchmarks. As LLMs push the frontier, they are evaluated on new, challenging benchmarks (Jimenez et al., 2024; Merrill et al., 2026; Rein et al., 2024; Yue et al., 2024; 2025; Center for AI Safety et al., 2026; Foundation, 2026), leaving behind older, saturated ones (Cobbe et al., 2021; Zellers et al., 2019; Wang et al., 2018; Sakaguchi et al., 2019) on which frontier models are now considered to perform nearly perfectly. Yet near-perfect is not perfect: Vendrow et al. (2024) showed that frontier LLMs, such as o1-2024-12-17 (high) (OpenAI, 2024), retain rare failures, around 0.1%, on GSM8K (Cobbe et al., 2021) even after correcting noisy label errors. Relatedly, GSM-Symbolic (Mirzadeh et al., 2025) demonstrates that frontier models’ near-perfect performance degrades on parameterized variants of the same problem templates. We focus on rigorous estimation of failure rates for models on highly saturated benchmarks, targeting regimes as extreme as “five-nines” (99.999%) accuracy, where the rarity of failures makes accurate estimation computationally costly compared to less saturated settings.

Efficient LLM evaluation. The autoregressive nature of LLMs makes inference expensive, and this cost grows as test-time compute scales (OpenAI, 2024; Guo et al., 2025). Several approaches reduce evaluation cost by selecting informative subsets of benchmark items (Maia Polo et al., 2024; Vivek et al., 2024; Perlitz et al., 2024; Hofmann et al., 2025), leveraging historical information (Wu et al., 2026), or using difficulty-adaptive testing (Truong et al., 2025). These studies target general capability evaluation on non-saturated benchmarks, whereas our work specifically addresses the rare-failure regime of highly saturated benchmarks.

Rare-event estimation with the CEM. The CEM (Rubinstein, 1999; Rubinstein & Kroese, 2004; De Boer et al., 2005) is a method for estimating rare-event probabilities by iteratively learning a proposal distribution concentrated on the rare-event region, enabling importance sampling with substantially reduced variance. A few works have adopted the CEM for rare-event estimation, including estimating failure probabilities of black-box physical systems in safety-critical applications (Arief et al., 2021) and estimating rare predictions of linear predictors (Bai et al., 2022). Our work is the first to apply the CEM to LLM evaluation, where the rare event is a majority-vote failure, the parameter space is a high-dimensional categorical distribution over problem instances, and each LLM inference is expensive.

3 Problem and Experimental Setup

Problem parameterization. To evaluate a model over a broad problem distribution, we construct a parameterized representation of task instances (e.g., mathematical reasoning prompts) as a mapping from the parameters Z to natural language prompts. Specifically, we define $Z = (Z_1, \dots, Z_d)$ as a vector of d independent categorical variables, where each Z_i takes values in a finite domain $\mathcal{Z}_i$. Following Mirzadeh et al. (2025), we parameterize all types of variables — including nouns and numerical values.

To illustrate, consider a template where specific nouns and values are treated as parameters:

Prompt (Template 0)

$\{Z_{\text{Name}}\}$ saw a $\{Z_x\}$ -foot $\{Z_{\text{Fish}}\}$ with $\{Z_y\}$ remoras, each $\{Z_z\}$ -inches long, attached to it. What percentage of the $\{Z_{\text{Fish}}\}$ ’s body length is the combined length of the remoras?

Each placeholder corresponds to a variable Z_i sampled from its domain $\mathcal{Z}_i$ (e.g., $Z_{\text{Name}} \in \mathcal{Z}_{\text{Name}}$, $Z_x \in \mathcal{Z}_x$). This formulation provides a structured, factorized representation of the problem input space. Let $\mathcal{Z} = \prod_{i=1}^d \mathcal{Z}_i$ denote the full parameter space. Details for templates are given in Appendix G.